As requested, I am acting as the Principal Data Architect and Enterprise Systems Designer to provide the definitive, exhaustive Physical Data Design for V1 of the DTE Platform.
This design strictly adheres to the provided chronological artifacts, incorporating the Ruthless Scope Cut, the V1 Domain Specification, and the critical modifications dictated by the Compliance Gap Analysis and the Regulatory Distribution Strategy. It relies on relational database paradigms, leveraging PostgreSQL-specific data types (e.g., UUID, JSONB, NUMERIC) where optimal.
Here is the complete physical data design.

SECTION 1 — Database Design Principles

To ensure absolute regulatory fidelity and multi-year sustainability, the physical design is bound by the following immutable principles:

- **Strict Separation of Software from Regulation:** Regulatory rules, MH catalogs, and schemas are strictly treated as data updates, never hardcoded into application execution paths.
- **The Snapshot Law:** A validated fiscal document is a photograph of facts. It copies master data (issuer, receptor, items, catalog values) entirely by value into immutable structures. Physical foreign keys between transactional records and live master data are strictly prohibited to prevent retroactive corruption.
- **Append-Only Evidence and Audit:** Historical truth is immutable. Evidence artifacts and audit entries are append-only; UPDATE and DELETE privileges must be physically revoked at the database layer for these tables.
- **Cryptographic Verifiability:** Audit trails utilize tamper-evident hash chaining. Evidence bundles persist cryptographic hashes of content computed at the exact moment of capture.
- **Effective Dating over Deletion:** Regulatory catalogs evolve through effective dating (valid_from, valid_to) rather than destructive updates. Old codes are deprecated, never deleted.
- **Mathematical Precision:** All financial values rely on exact-scale numerics (NUMERIC), completely eliminating floating-point reconciliation errors.

SECTION 2 — Physical Entity Inventory

Every persistent structure required for V1 is listed below with its strict operational parameters.

Issuance Context

- fiscal_document: The central transactional root representing a DTE. **Ownership**: Issuance. **Lifecycle**: Draft $\rightarrow$ Validated $\rightarrow$ Signed $\rightarrow$ (Submitted | Contingency-Held) $\rightarrow$ (Accepted | Rejected | Invalidated). **Retention**: Indefinite.
- document_line: A single billed good/service. **Ownership**: Issuance. **Lifecycle**: Frozen at document signing. **Retention**: Indefinite.
- control_number_series: Single authority for gapless correlatives. **Ownership**: Issuance. **Lifecycle**: Monotonic incrementation. **Retention**: Indefinite.

- transmission_record: Exhaustive lifecycle tracker for authority interactions. **Ownership**: Issuance. **Lifecycle**: Append-only. **Retention**: Indefinite.
- transmission_attempt: A single API payload/response round trip. **Ownership**: Issuance. **Lifecycle**: Immutable upon creation. **Retention**: Indefinite.

Governance Contexts (Invalidation & Contingency)

- invalidation_event: Executes lawful voiding of accepted documents. **Ownership**: Invalidation. **Lifecycle**: Draft $\rightarrow$ Submitted $\rightarrow$ (Accepted | Rejected). **Retention**: Indefinite.
- contingency_episode: Governs bounded offline issuance periods. **Ownership**: Contingency. **Lifecycle**: Open $\rightarrow$ Reconciling $\rightarrow$ Closed. **Retention**: Indefinite.

Master Data Context (Parties & Catalog)

- counterparty: Reusable receptor master data. **Ownership**: Parties. **Lifecycle**: Mutable/Soft-delete. **Retention**: Retained if active.
- item: Reusable product master data. **Ownership**: Catalog. **Lifecycle**: Mutable/Soft-delete. **Retention**: Retained if active.

Evidence & Audit Contexts

- evidence_bundle: Immutable proof vault mapping 1:1 to signed documents. **Ownership**: Evidence. **Lifecycle**: Append-only. **Retention**: Full legal horizon.
- evidence_artifact: Cryptographically hashed payloads, stamps, and representations. **Ownership**: Evidence. **Lifecycle**: Immutable. **Retention**: Full legal horizon.
- audit_trail: Global container for the hash chain. **Ownership**: Audit. **Lifecycle**: Append-only. **Retention**: Indefinite.
- audit_entry: A single tamper-evident action record. **Ownership**: Audit. **Lifecycle**: Immutable. **Retention**: Indefinite.

Configuration & Access Context

- app_user: Human actors. **Ownership**: Access. **Lifecycle**: Mutable. **Retention**: Retained.
- issuer_configuration: Singleton definition of the legal entity. **Ownership**: Configuration. **Lifecycle**: Mutable. **Retention**: Retained.
- establishment: Physical locations mapped to MH catalogs. **Ownership**: Configuration. **Lifecycle**: Mutable. **Retention**: Retained.
- point_of_sale: Logical issuance generators mapping to control sequences. **Ownership**: Configuration. **Lifecycle**: Mutable. **Retention**: Retained.

Regulatory & Type Registry Context

- mh_catalog: Defines an MH-mandated list. **Ownership**: System. **Lifecycle**: Append-only. **Retention**: Indefinite.
- catalog_entry: Effective-dated options within a catalog. **Ownership**: System. **Lifecycle**: Append-only. **Retention**: Indefinite.

- schema_version_binding: Structural constraints for a DTE Type at a point in time. **Ownership**: Type Registry. **Lifecycle**: Append-only. **Retention**: Indefinite.
- regulatory_bundle: Cryptographically verified payload dictating compliance. **Ownership**: Distribution. **Lifecycle**: Immutable. **Retention**: Indefinite.

SECTION 3 — Table Definitions

Note: Foreign keys omitted conceptually across Bounded Contexts per Domain Driven Design rules, but physical UUID types are enforced. JSONB is used extensively for snapshots. NUMERIC(18,4) enforces V1 financial precision standards.

Table: fiscal_document

Purpose: Core DTE transaction record.
Primary Key: document_id (UUID)

Column Name	Data Type	Nullable	Default	Constraints/Notes
document_id	UUID	No	gen_random_uuid()	Primary Key.
dte_type	VARCHAR(2)	No	-	'01' or '03'.
schema_version	VARCHAR(10)	Yes	-	Pinned at validation.
generation_code	UUID	Yes	-	Globally unique.
control_number	VARCHAR(31)	Yes	-	Format: DTE-XX-XXXXXXXX-NNN....
environment_code	VARCHAR(2)	Yes	-	Cat-002: 00 or 01.
transmission_model_code	VARCHAR(1)	Yes	-	Cat-003: 1 or 2.
operation_type	VARCHAR(1)	Yes	-	Cat-004: 1 or 2.

_code				
issue_date	DATE	Yes	-	MH parsed calendar date.
issue_time	TIME	Yes	-	MH parsed exact time.
status	VARCHAR(30)	No	'DRAFT'	Draft, Validated, Signed....
issuer_snapshot	JSONB	Yes	-	Frozen at validation.
receptor_snapshot	JSONB	Yes	-	Frozen at validation.
tax_summary	JSONB	Yes	-	Array of computed taxes.
net_total	NUMERIC(18,4)	Yes	-	Sum before taxes.
tax_total	NUMERIC(18,4)	Yes	-	Sum of taxes.
gross_total	NUMERIC(18,4)	Yes	-	Payable amount.
sub_total_sales	NUMERIC(18,4)	Yes	-	Pre-discount line sum.
global_discounts	NUMERIC(18,4)	Yes	0.0000	Global discount buckets.
retained_taxes	NUMERIC(18,4)	Yes	0.0000	IvaRete1/IvaPerci1.
total_in_letters	TEXT	Yes	-	Letras representation.

payment_forms	JSONB	Yes	-	Array of PaymentAllocations. *(Replaces* operation_condition*).*
contingency_episode_id	UUID	Yes	-	Links if offline issuance.
created_at	TIMESTAMPTZ	No	NOW()	Internal timestamp.
updated_at	TIMESTAMPTZ	No	NOW()	Internal timestamp.

Table: document_line

Purpose: Billed line items linked to a document.
Primary Key: line_id (UUID)

Column Name	Data Type	Nullable	Default	Constraints/Notes
line_id	UUID	No	gen_random_uuid()	Primary Key.
document_id	UUID	No	-	FK to fiscal_document.
line_number	INTEGER	No	-	Sequential 1, 2, 3.
item_snapshot	JSONB	No	-	Includes item_type_code (Cat-011).
quantity	NUMERIC(18,4)	No	-	Must be > 0.

unit_price	NUMERIC(18,4)	No	-	Must be >= 0.
discount_amount	NUMERIC(18,4)	No	0.0000	Line level discount.
tax_bucket	VARCHAR(20)	No	-	Gravada, Exenta, NoSujeta.
tax_lines	JSONB	No	-	Specific taxes computed.
line_total	NUMERIC(18,4)	No	-	Qty * Price + Taxes.

Table: control_number_series

Purpose: Strict, lockable sequence generator.
Primary Key: series_id (UUID)

Column Name	Data Type	Nullable	Default	Constraints/Notes
series_id	UUID	No	gen_random_uuid()	Primary Key.
dte_type	VARCHAR(2)	No	-	e.g., '01', '03'.
establishment_code	VARCHAR(4)	No	-	Prefix scope.
pos_code	VARCHAR(4)	No	-	Suffix scope.
next_correlative	BIGINT	No	1	Strictly increments.

Table: invalidation_event

Purpose: Tracking plain void commands.
Primary Key: invalidation_id (UUID)

Column Name	Data Type	Nullable	Default	Constraints/Notes
invalidation_id	UUID	No	gen_random_uuid()	Primary Key.
target_document_id	UUID	No	-	Original FiscalDocument reference.
reason_type	VARCHAR(20)	No	-	'Rescind' or 'Other'.
justification	TEXT	No	-	Human rationale.
requester_identity	JSONB	No	-	MH mandated persona.
executor_identity	JSONB	No	-	MH mandated persona.
transmission_record_id	UUID	Yes	-	Link to comms history.
status	VARCHAR(30)	No	'DRAFT'	Draft, Submitted, Accepted, Rejected.

Table: contingency_episode

Purpose: Manage offline issuance windows.
Primary Key: episode_id (UUID)

Column Name	Data Type	Nullable	Default	Constraints/Notes
episode_id	UUID	No	gen_random_uuid()	Primary Key.

opened_at	TIMESTAMPTZ	No	-	Required.
closed_at	TIMESTAMPTZ	Yes	-	Settled.
reason_code	VARCHAR(30)	No	-	MH unavailability code.
declarant_identity	JSONB	No	-	MH mandated persona.
status	VARCHAR(30)	No	'OPEN'	Open, Reconciling, Closed, WindowBreached.
notification_tx_id	UUID	Yes	-	Notification transmission.

Table: regulatory_bundle

Purpose: Stores cryptographic payloads from the vendor.
Primary Key: bundle_version (VARCHAR)

Column Name	Data Type	Nullable	Default	Constraints/Notes
bundle_version	VARCHAR(50)	No	-	Format: YYYY.MM.DD.Rev. PK.
digital_signature	TEXT	No	-	Signature preventing tampering.
min_platform_version	VARCHAR(20)	No	-	Semantic Version constraint.
applied_at	TIMESTAMPTZ	No	NOW()	When staged/applied

				.
status	VARCHAR(20)	No	'ACTIVE'	Active, Superseded, RolledBack.

Table: mh_catalog & catalog_entry

Purpose: MH validation registries with effective dating.
mh_catalog (Primary Key: catalog_identifier)

Column	Type	Null	Default	Notes
catalog_identifier	VARCHAR(20)	No	-	e.g., 'Cat-011'. PK.
name	VARCHAR(100)	No	-	Human readable.
description	TEXT	Yes	-	Regulatory purpose.

catalog_entry (Primary Key: entry_id UUID)

Column	Type	Null	Default	Notes
entry_id	UUID	No	gen_random_uuid()	PK.
catalog_identifier	VARCHAR(20)	No	-	FK to mh_catalog.
code	VARCHAR(20)	No	-	MH string value.
value	TEXT	No	-	Meaning.
valid_from	DATE	No	-	Effective dating.

valid_to	DATE	Yes	-	Null = forever.
status	VARCHAR(20)	No	'ACTIVE'	Active, Deprecated, Future.
metadata	JSONB	Yes	-	Extensible constraints.

SECTION 4 — Relationships

The core principle here is the removal of live master data constraints from historical artifacts.

- **fiscal_document (1) to document_line (N):**
  - *Parent:* fiscal_document. *Child:* document_line.
  - *Delete:* CASCADE if parent is Draft. Application logic strictly forbids deletion if Status > Draft.
  - *Update:* Restrict updates if Status > Draft.
- **fiscal_document (1) to evidence_bundle (1):**
  - *Parent:* fiscal_document. *Child:* evidence_bundle.
  - *Ownership:* evidence_bundle points back to fiscal_document.
  - *Delete:* RESTRICT. Evidence must never be deleted.
- **mh_catalog (1) to catalog_entry (N):**
  - *Parent:* mh_catalog. *Child:* catalog_entry.
  - *Delete:* RESTRICT. Catalogs are append-only.
- **fiscal_document to counterparty / item (None):**
  - There is **zero** physical foreign key relationship between transactional documents and master data. All historical data relies entirely on the JSONB snapshots embedded within the document tables to ensure retroactive corruption is physically impossible.

SECTION 5 — Constraints

Physical layer guards preventing bad data irrespective of application layer flaws.

- **Unique Constraints:**
  - idx_uq_control_number: UNIQUE (dte_type, establishment_code, pos_code, next_correlative) on control_number_series guarantees gapless non-collision.
  - idx_uq_generation_code: UNIQUE (generation_code) on fiscal_document.
  - idx_uq_bundle_document: UNIQUE (document_id) on evidence_bundle.
- **Check Constraints:**
  - chk_positive_money: CHECK (quantity > 0 AND unit_price >= 0) on document_line.
  - chk_net_total_positive: CHECK (net_total >= 0) on fiscal_document.
- **Business Constraints (Application + DB triggers):**
  - **Immutable Signed Documents:** PostgreSQL Triggers BEFORE UPDATE OR DELETE ON fiscal_document blocking any modification if status IN ('Signed', 'Submitted', 'Accepted',

'Rejected', 'Invalidated').
  - **Audit Trail Chain Check:** Triggers on audit_entry rejecting inserts if previous_hash does not strictly equal the current_hash of sequence_number - 1.

SECTION 6 — Snapshot Persistence Model

Master data is copied by-value at the moment of validation.

- **Physical Columns:** Stored as JSONB fields (issuer_snapshot, receptor_snapshot, item_snapshot) directly inside the transactional tables.
- **Immutability Strategy:** Once written, updating the root aggregate (fiscal_document) marks the fields read-only via trigger.
- **Schema Enforcement:**
  - issuer_snapshot: Enforces schema containing NIT, NRC, EconomicActivityCode, EstablishmentTypeCode, and strict ContactInfo (phone/email).
  - receptor_snapshot: Enforces schema capturing TaxIdentityString, EconomicActivityCode, EconomicActivityDescription, CommercialName, and ContactInfo.
  - item_snapshot: Embeds description, unit_of_measure, item_type_code (goods vs services, Cat-011), and default tax behaviors.

SECTION 7 — Audit Persistence Model

The physical implementation of the hash-chained ledger.

Table: audit_entry

Column	Type	Null	Default	Notes
sequence_number	BIGSERIAL	No	-	Monotonically increasing PK.
timestamp	TIMESTAMPTZ	No	NOW()	Exact server time.
actor_identity	UUID	No	-	Link to app_user.
actor_role	VARCHAR(30)	No	-	Owner/Biller.
action_type	VARCHAR(50)	No	-	What happened.
target_identity	UUID	No	-	Subject ID.
outcome_statu	VARCHAR(20)	No	-	Success,

s				Failure, Denied.
previous_hash	VARCHAR(64)	No	-	SHA-256 of previous block.
current_hash	VARCHAR(64)	No	-	Generated hash.

- **Persistence Strategy:** Inserts exclusively. A DB-level role is utilized by the application that only holds INSERT privileges. Deletes and updates are blocked at the PostgreSQL permission layer.

SECTION 8 — Evidence Persistence Model

An incontrovertible vault of proofs mapping 1:1 to a document.
Table: evidence_artifact

Column	Type	Null	Default	Notes
artifact_id	UUID	No	gen_random_uuid()	PK.
bundle_id	UUID	No	-	FK to evidence_bundle.
artifact_type	VARCHAR(30)	No	-	Payload, Stamp, PDF, AuthResponse.
captured_at	TIMESTAMPTZ	No	NOW()	Capture timestamp.
content_blob	BYTEA	No	-	Binary payload (JSON or PDF).
content_hash	VARCHAR(64)	No	-	SHA-256 digest calculated on insert.

- **Storage Strategy:** BYTEA guarantees the artifact is transactionally locked inside the database backup, satisfying self-containment requirements for regulatory survival.

SECTION 9 — Catalog Persistence Model

Designed to absorb Ministry of Finance updates asynchronously without modifying code.

- **Versioning:** Data driven solely by valid_from and valid_to in the catalog_entry table.
- **Resolution:** When the Issuance context requests catalog validity (e.g., verifying a tax code), it queries catalog_entry where issue_date BETWEEN valid_from AND COALESCE(valid_to, 'infinity').
- **Deprecation:** Updating a catalog to deprecate an item simply sets valid_to to the mandate date and updates status = 'Deprecated'. The data remains intact to satisfy historical proofs.

SECTION 10 — Regulatory Bundle Persistence Model

The regulatory_bundle table receives signed packages from the vendor.

- **Activation:** The application reads the highest chronologically valid bundle_version where status = 'ACTIVE' to configure system rules.
- **Rollback/Failure:** If a bundle transaction fails midway, the bundle is discarded. If vendor logic issues a corrected bundle, the higher version number gracefully supersedes the old.
- **Compatibility Matrix:** min_platform_version strictly asserts whether the current binary can legally consume the bundle.

SECTION 11 — Index Strategy

- idx_fiscal_doc_status: CREATE INDEX ON fiscal_document (status) - Critical for polling jobs finding Signed documents to move to Submitted via BullMQ.
- idx_fiscal_doc_dates: CREATE INDEX ON fiscal_document (issue_date) - Critical for tax-book export reporting bounds.
- idx_catalog_effective: CREATE INDEX ON catalog_entry (catalog_identifier, valid_from, valid_to) - Accelerates point-in-time point catalog lookups during document validation.
- idx_evidence_bundle: CREATE INDEX ON evidence_artifact (bundle_id) - Ensures instantaneous retrieval of compliance packages upon auditor request.
- idx_audit_chain: CREATE UNIQUE INDEX ON audit_entry (sequence_number, previous_hash) - Physically enforces the chronological integrity constraint at the B-tree level.

SECTION 12 — Concurrency Strategy

- **Control Number Allocation:** Highly contested in automated setups. Uses PostgreSQL SELECT ... FOR UPDATE directly on the control_number_series row. This blocks concurrent issuance threads for a specific POS just long enough to allocate the integer and commit, preventing gaps and collisions entirely.
- **Document Locking:** When an actor transitions a document to Validated, a database row-level lock ensures no concurrent actor alters the lines or totals while snapshots are capturing.
- **Distributed Queueing:** BullMQ (via Redis) handles outgoing transmissions to the MH. Redis locks enforce idempotency, ensuring the database is updated with outcomes precisely once without race conditions.

SECTION 13 — Archival Strategy

To support a 10+ year survival horizon without choking the active transaction engine.

- **Partitioning:** Native PostgreSQL table partitioning. fiscal_document and evidence_artifact partitioned dynamically by issue_date RANGE (YEAR).
- **Historical Access Model:** Active database queries transparently target partitions. Years > 3 map to read-only tablespaces pushed to cheaper underlying block storage.
- **Evidence Retention:** Because evidence relies strictly on BYTEA blobs and hashes, dropping an old counterparty row from the active table has zero impact. The partitions remain fully sufficient to prove compliance independently.

SECTION 14 — Performance Considerations

- **Denormalization via JSONB:** Validating the fiscal_document compiles all relevant data into JSONB. The representation rendering service or tax-book export script never has to execute 5-way joins across counterparty and item tables. It simply reads the fiscal_document row.
- **Blob Segregation:** BYTEA column content_blob on evidence_artifact uses PostgreSQL TOAST storage. This keeps the primary table lean, ensuring SELECT * FROM evidence_artifact queries for metadata scanning remain fast even when containing millions of 2MB PDFs.